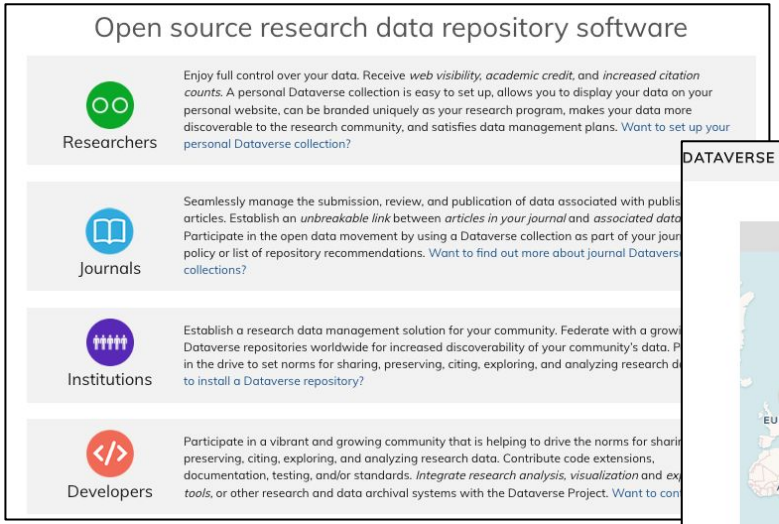
Permissions on a Dataverse Supported Repository

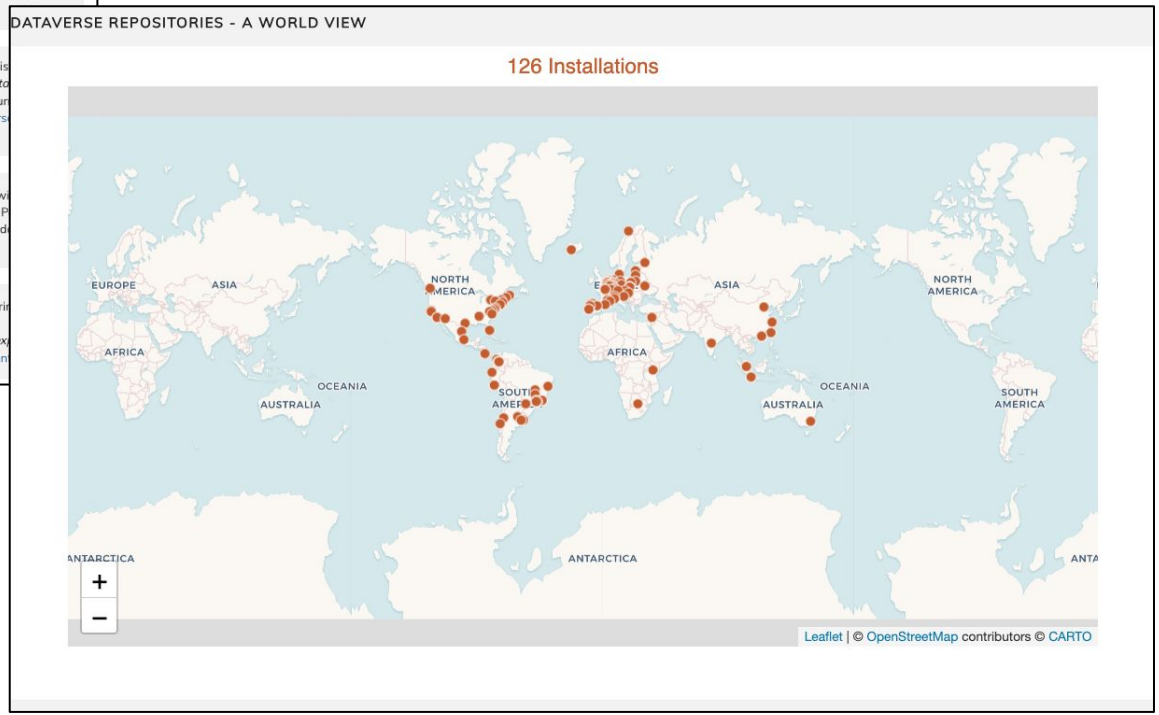
Sonia Barbosa and Gustavo Durand

The Dataverse® Project

The Dataverse Project - Overview

The Dataverse Project





Dataverse Repository Features

The Dataverse® Project

- Support for FAIR Data Principles
- Data citation for datasets and files
- OAI-PMH (Harvesting)
- APIs for interoperability and custom integrations
- API client libraries
- DataCite integration
- Login via Shibboleth
- Login via ORCID, Google, GitHub, or Microsoft
- Login via OpenID Connect (OIDC)
- Internationalization
- Versioning
- Restricted files
- Embargo
- Custom licenses
- Custom terms of use
- Publishing workflow support
- File hierarchy
- File previews
- Preview and analysis of tabular files
- Usage statistics and metrics
- Guestbook
- Fixity checks for files
- File download in R and TSV format
- Faceted search
- Customization of collections
- Private URL
- Widgets
- Notifications
- Schema.org JSON-LD
- External tools
- External vocabulary
- Dropbox integration
- GitHub integration
- Integration with Jupyter notebooks
- User management
- Curation status labels
- Branding
- Backend storage on S3 or Swift
- Direct upload and download for S3
- Export data in BagIt format
- Post-publication automation (workflows)
- Pull header metadata from Astronomy (FITS) files
- Provenance
- Support for rsync
- Auxiliary files for data files

https://dataverse.org/software-features

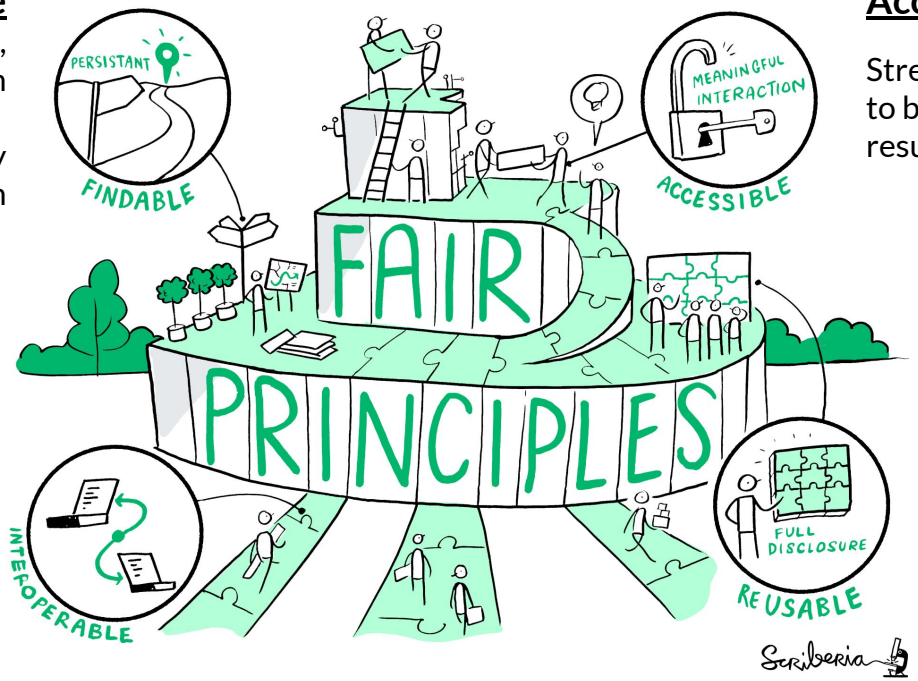
The Dataverse Project and FAIR

FAIR Principles and Dataverse

Findable

Increases visibility, citations, and impact of research

Supports knowledge discovery and innovation



Accessible

Streamlines and maximizes ability to build upon previous research results

Interoperable

Supports and promotes inter- and cross-disciplinary data and reuse

Reusable

Promotes use and reuse of data allowing resources to be allocated wisely

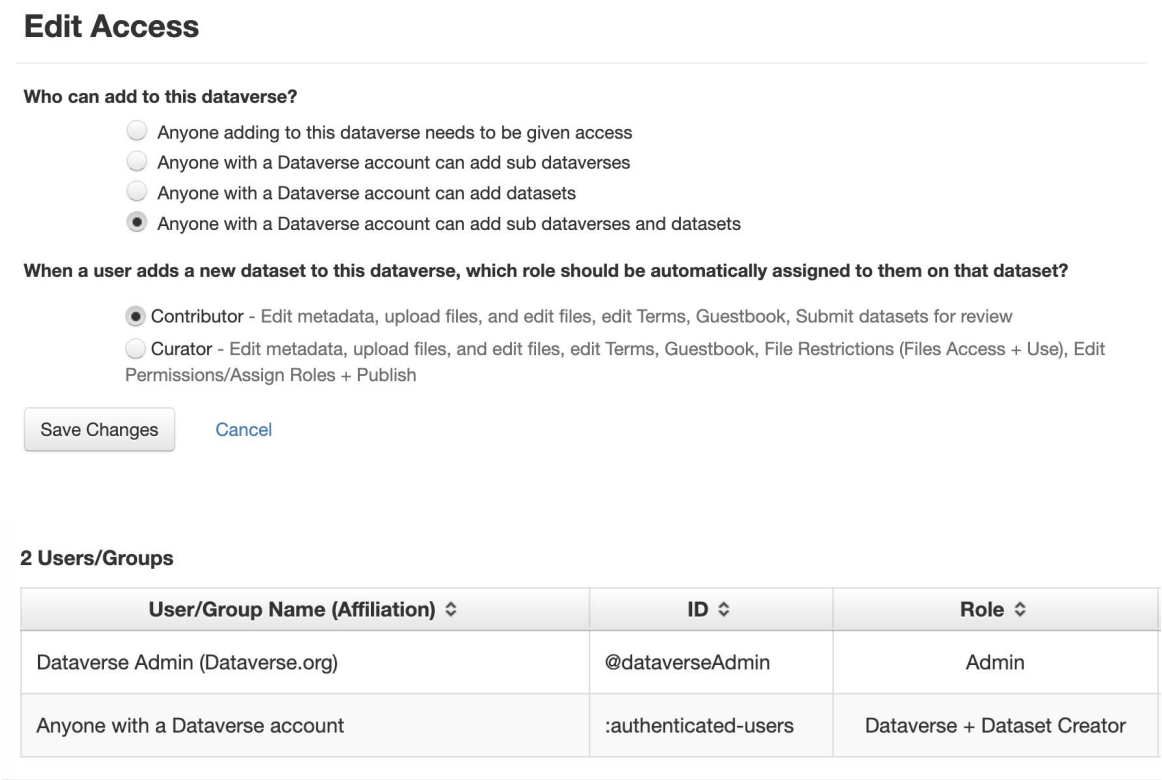
Improves reproducibility and reliability of research results

Wilkinson et al. 2016. [The FAIR Guiding Principles for scientific data management and stewardship]

Technical Overview of Permissions

Flexible Permission System

- Supports multiple workflows by controlling who can add to your Dataverse collection, what they can, and what role they have on and created Datasets
- **Roles are defined as a set of permissions to grant to users or to groups**
- Groups can be defined statically or dynamically (e.g. users logging in from the same institution, via Shibboleth)



Technical Design

An application design that enforces permission-based policies

- Supports declarative syntax for stating “Action X on object Y requires permissions $P_1..P_n$”
  - With dynamic overrides when needed
- based on an adaptation of the classic Command Pattern

Adapting the Command Pattern

- **Commands** specify:
  - which permissions are required to execute them,
  - which model objects are involved
  - what is the context in which the command was submitted
- A **central command execution engine** that can verify that a command should indeed be executed prior to invoking it.
- Circumvents the need for sprinkling multiple permission checks at various places in the code

Adapting the Command Pattern

This design ensures that:

- Users are only able to perform actions they have permissions to perform
- Developers must explicitly state which permissions are required to run each command
- To the extent that static permissions declaration is used, the permissions required to run each command are *declaratively* documented in the code

Sample Command: Get Draft Dataset Version

```
@RequiredPermissions( Permission.ViewUnpublishedDataset )
public class GetDraftDatasetVersionCommand extends AbstractCommand<DatasetVersion>{

    private final Dataset ds;

    public GetDraftDatasetVersionCommand(DataverseRequest aRequest, Dataset
anAffectedDataset) {
        super(aRequest, anAffectedDataset);
        ds = anAffectedDataset;
    }

    @Override
    public DatasetVersion execute(CommandContext ctxt) throws CommandException {
        return ds.getOrCreateEditVersion();
    }
```

Dynamic Permissions

Permissions can be defined at runtime:

```
@RequiredPermissions({})
public class GetDatasetCommand extends AbstractCommand<Dataset>{

  . . .

  @Override
  public Map<String,Set <Permissions>> getRequiredPermissions(){({
      if ( dataset.isReleased() ) {
          return Collections.emptySet();
      } else  {
          return Collections.singelton(Permission.ViewUnpublishedDataset));
      }
  }
}
```

/!\ Code cleaned for clarity

Command Composition

Commands can be composed of other commands:

```
@RequiredPermissions({})
public class GetLatestAccessibleDatasetVersionCommand extends AbstractCommand<DatasetVersion>{
  private final Dataset ds;

  . . .

  @Override
  public DatasetVersion execute(CommandContext ctxt) throws CommandException {
      try {
          return ctxt.engine().submit(new GetDraftDatasetVersionCommand(getRequest(), ds));
      } catch(PermissionException ex) {}
          return ctxt.engine().submit(
                          new GetLatestPublishedDatasetVersionCommand(getRequest(),ds));
      }
  }
}
```

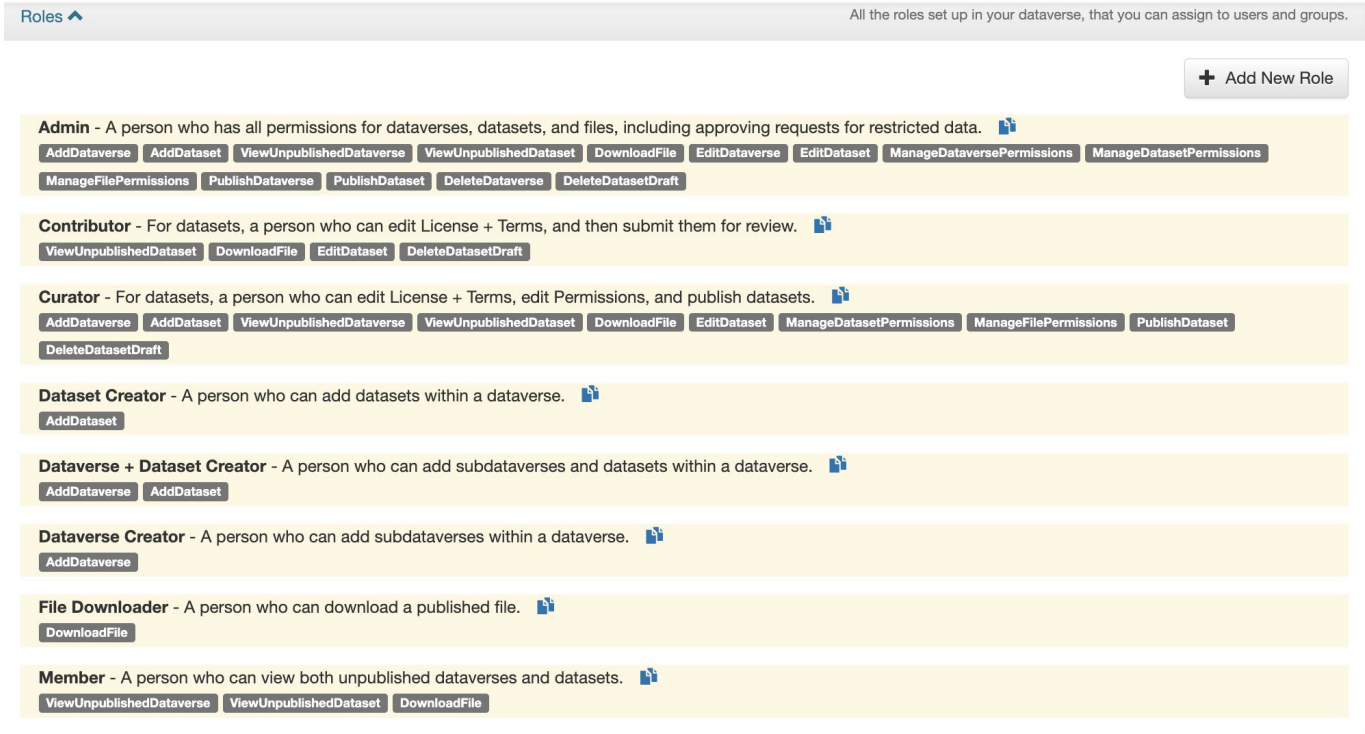
/!\ Code cleaned for clarity

Permissions (by type of operation)

- **Create**
  - AddDataverse
  - AddDataset
- **Read**
  - ViewUnpublishedDataverse
  - ViewUnpublishedDataset
  - DownloadFile
- **Update**
  - EditDataverse
  - EditDataset
  - ManageDataversePermissions
  - ManageDatasetPermissions
  - ManageFilePermissions
  - PublishDataverse
  - PublishDataset
- **Delete**
  - DeleteDataverse
  - DeleteDatasetDraft

Roles

A **Role** is a collection of Permissions:



Permissions (by scope)

- **Dataverse Collection**
  - AddDataverse
  - AddDataset
  - ViewUnpublishedDataverse
  - EditDataverse
  - ManageDataversePermissions
  - PublishDataverse
  - PublishDataset*
  - DeleteDataverse
- **Dataset**
  - ViewUnpublishedDataset
  - EditDataset
  - ManageDatasetPermissions
  - PublishDataset*
  - DeleteDatasetDraft
- **File**
  - DownloadFile
  - ManageFilePermissions

Role Inheritance

Roles **<u>are</u>** inherited:

- From **dataverse collections** to **datasets**
- From **datasets** to **files**

_____

Roles **<u>are not</u>** inherited:

- From **dataverse collections** to **dataverse subcollections**
- Note, there is an API to add dataverse collection role assignments to dataverse subcollections (superuser only)

Inheritance Examples

File Downloader role at the dataset level

- User has the permission to download ALL files in that dataset
- This includes future files that are uploaded to the dataset

_____

Curator role at the dataverse collection level

- User has the permission to edit, publish, etc. ALL datasets in that dataverse collection
- This includes future datasets that are added to the dataverse collection

Permissions in the User Interface

Roles

All the roles set up in your dataverse, that you can assign to users and groups.

+ Add New Role

Admin - A person who has all permissions for dataverses, datasets, and files, including approving requests for restricted data.

AddDataverse AddDataset ViewUnpublishedDataverse ViewUnpublishedDataset DownloadFile EditDataverse EditDataset ManageDataversePermissions ManageDatasetPermissions ManageFilePermissions PublishDataverse PublishDataset DeleteDataverse DeleteDatasetDraft

Contributor - For datasets, a person who can edit License + Terms, and then submit them for review.

ViewUnpublishedDataset DownloadFile EditDataset DeleteDatasetDraft

Curator - For datasets, a person who can edit License + Terms, edit Permissions, and publish datasets.

AddDataverse AddDataset ViewUnpublishedDataverse ViewUnpublishedDataset DownloadFile EditDataset ManageDatasetPermissions ManageFilePermissions PublishDataset DeleteDatasetDraft

Dataset Creator - A person who can add datasets within a dataverse.

AddDataset

Dataverse + Dataset Creator - A person who can add subdataverses and datasets within a dataverse.

AddDataverse AddDataset

Dataverse Creator - A person who can add subdataverses within a dataverse.

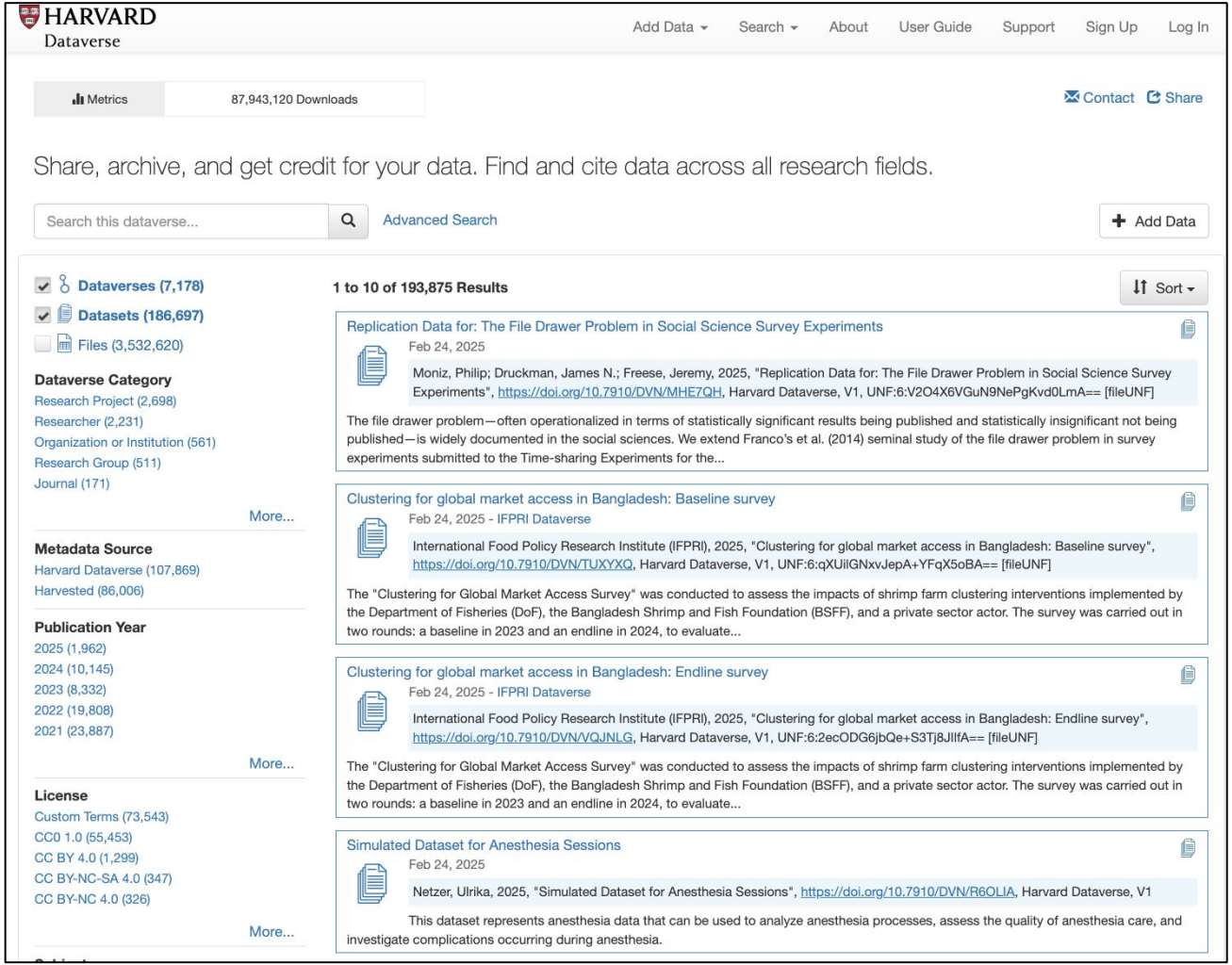
AddDataverse

File Downloader - A person who can download a published file.

DownloadFile

Member - A person who can view both unpublished dataverses and datasets.

ViewUnpublishedDataverse ViewUnpublishedDataset DownloadFile



Open to all for account creation to create and publish a collection and deposit datasets

Collections can set up their own controlled workflow to allow the same or use other roles and workflow options such as “SUBMIT FOR REVIEW” workflow

Account required for access to all restricted data

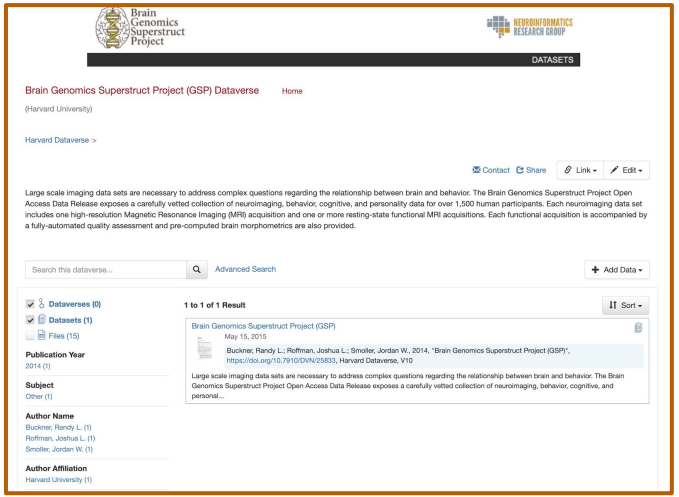
Collection-level

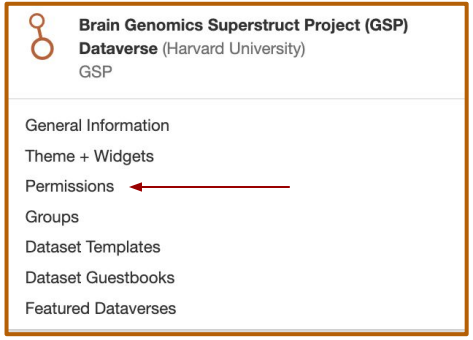
Current Features

Collections

- Own administration
- Own branding (and can be embedded anywhere)

- Who can add to my collection
- Who can see draft content
- Who can delete my collection
- Who can publish to my collection

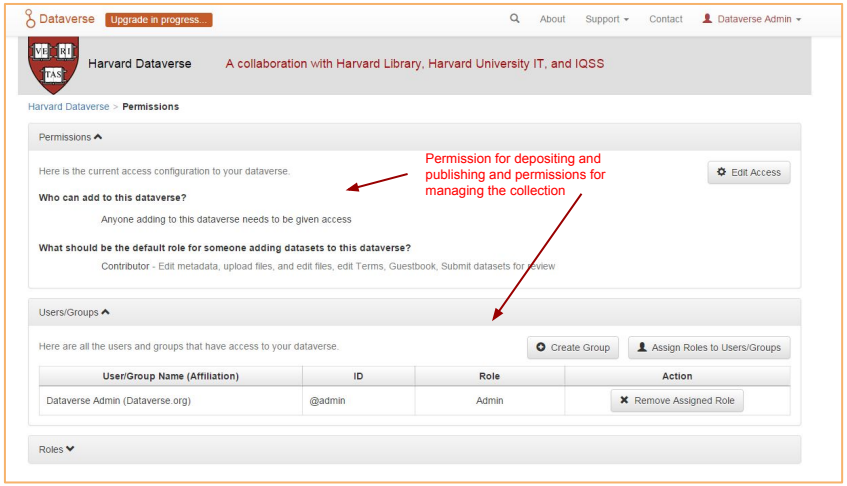


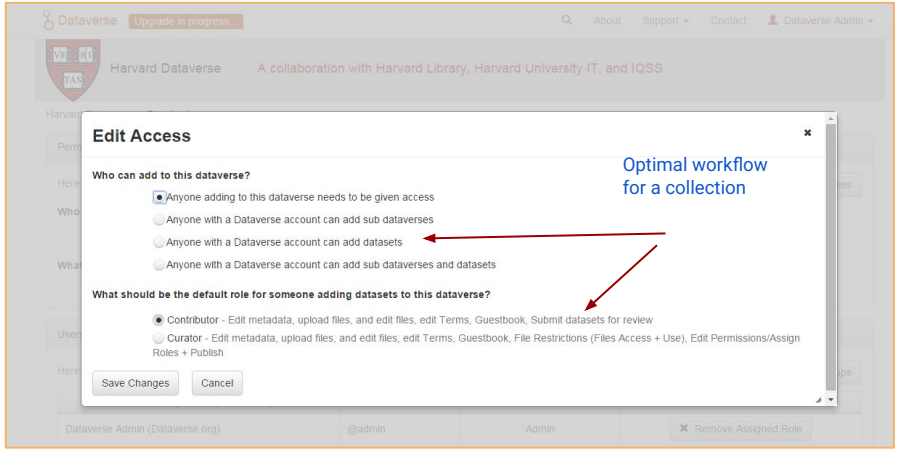


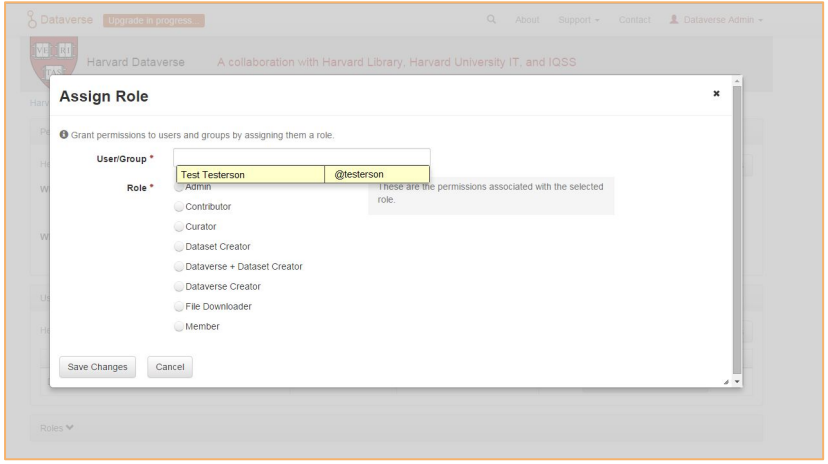
Supported Metadata

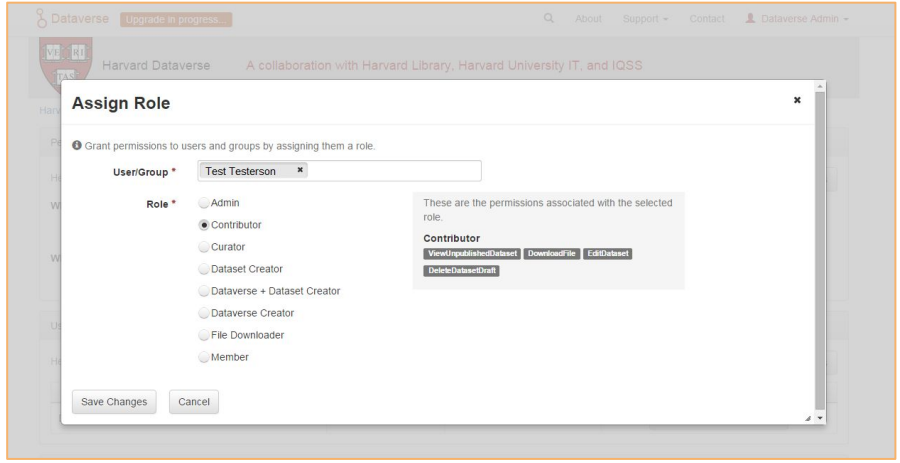
Detailed below are what metadata schemas we support for Citation and Domain Specific Metadata in the Dataverse Project:

- Citation Metadata (see .tsv version): compliant with DDI Lite, DDI 2.5 Codebook, DataCite 3.1, and Dublin Core's DCMI Metadata Terms . Language field uses ISO 639-1 controlled vocabulary.
- Geospatial Metadata (see .tsv version): compliant with DDI Lite, DDI 2.5 Codebook, DataCite, and Dublin Core. Country / Nation field uses ISO 3166-1 controlled vocabulary.
- Social Science & Humanities Metadata (see .tsv version): compliant with DDI Lite, DDI 2.5 Codebook, and Dublin Core.
- Astronomy and Astrophysics Metadata (see .tsv version): These metadata elements can be mapped/exported to the International Virtual Observatory Alliance's (IVOA) VOResource Schema format and is based on Virtual Observatory (VO) Discovery and Provenance Metadata.
- Life Sciences Metadata (see .tsv version): based on ISA-Tab Specification, along with controlled vocabulary from subsets of the OBI Ontology and the NCBI Taxonomy for Organisms.
- Journal Metadata (see .tsv version): based on the Journal Archiving and Interchange Tag Set, version 1.2.


Dataverse
Upgrade in progress...
About
Support
Contact
Dataverse Admin
Harvard Dataverse
A collaboration with Harvard Library, Harvard University IT, and IQSS
Harvard Dataverse > Permissions
Permissions
Here is the current access configuration to your dataverse.
Edit Access
Who can add to this dataverse?
Anyone adding to this dataverse needs to be given access
What should be the default role for someone adding datasets to this dataverse?
Contributor - Edit metadata, upload files, and edit files, edit Terms, Guestbook, Submit datasets for review
Permission for depositing and publishing and permissions for managing the collection
Users/Groups
Here are all the users and groups that have access to your dataverse.
Create Group
Assign Roles to Users/Groups
User/Group Name (Affiliation)
ID
Role
Action
Dataverse Admin (Dataverse.org)
@admin
Admin
Remove Assigned Role
Roles


Edit Access
Who can add to this dataverse?
Anyone adding to this dataverse needs to be given access
Anyone with a Dataverse account can add sub dataverses
Anyone with a Dataverse account can add datasets
Anyone with a Dataverse account can add sub dataverses and datasets
What should be the default role for someone adding datasets to this dataverse?
Contributor - Edit metadata, upload files, and edit files, edit Terms, Guestbook, Submit datasets for review
Curator - Edit metadata, upload files, and edit files, edit Terms, Guestbook, File Restrictions (Files Access + Use), Edit Permissions/Assign Roles + Publish
Save Changes
Cancel
Optimal workflow for a collection


Assign Role
Grant permissions to users and groups by assigning them a role.
User/Group *
Test Testerson
@testerson
Role *
Admin
Contributor
Curator
Dataset Creator
Dataverse + Dataset Creator
Dataverse Creator
File Downloader
Member
These are the permissions associated with the selected role.
Save Changes
Cancel


Assign Role
Grant permissions to users and groups by assigning them a role.
User/Group *
Test Testerson
Role *
Admin
Contributor
Curator
Dataset Creator
Dataverse + Dataset Creator
Dataverse Creator
File Downloader
Member
These are the permissions associated with the selected role.
Contributor
ViewUnpublishedDataset
DownloadFile
EditDataset
DeleteDatasetDraft
Save Changes
Cancel

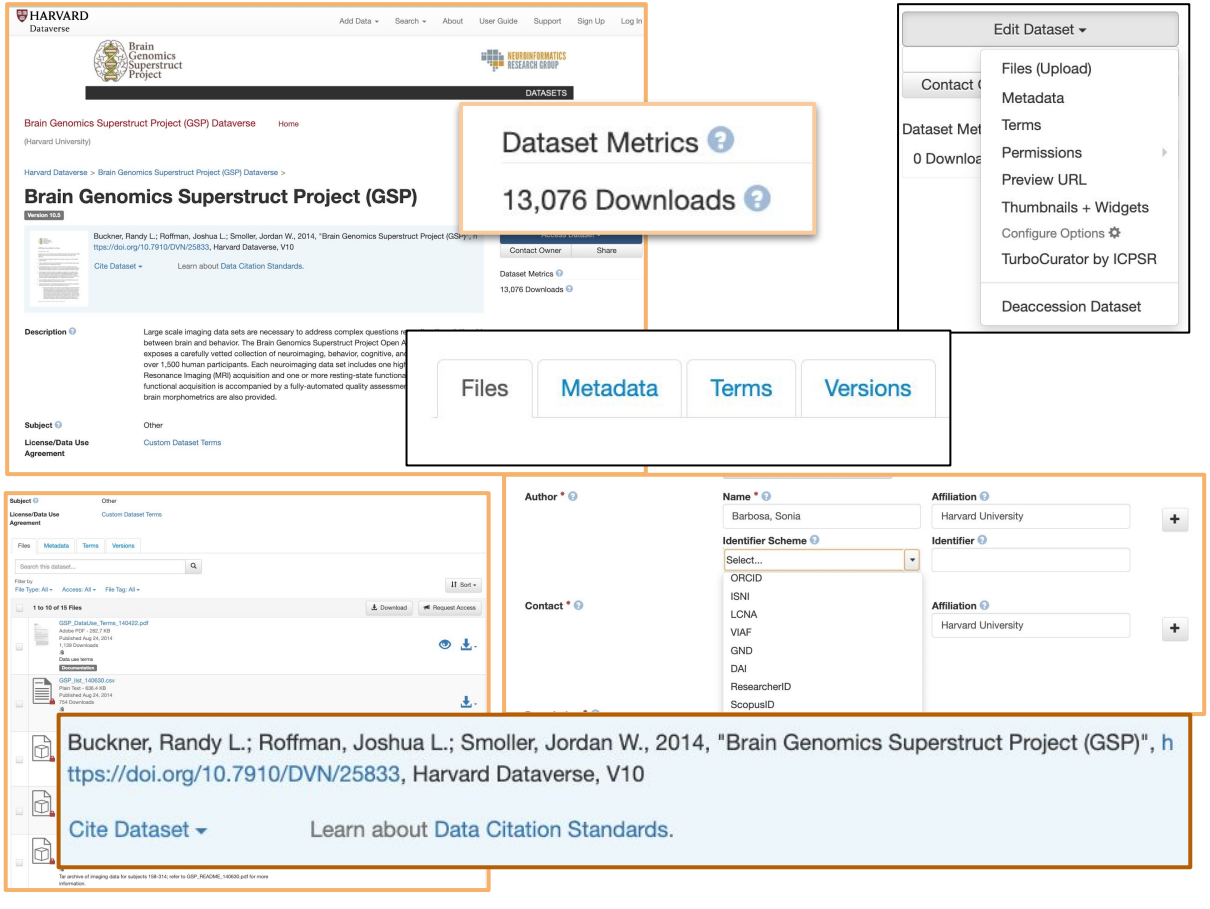
Dataset-level

Current Features

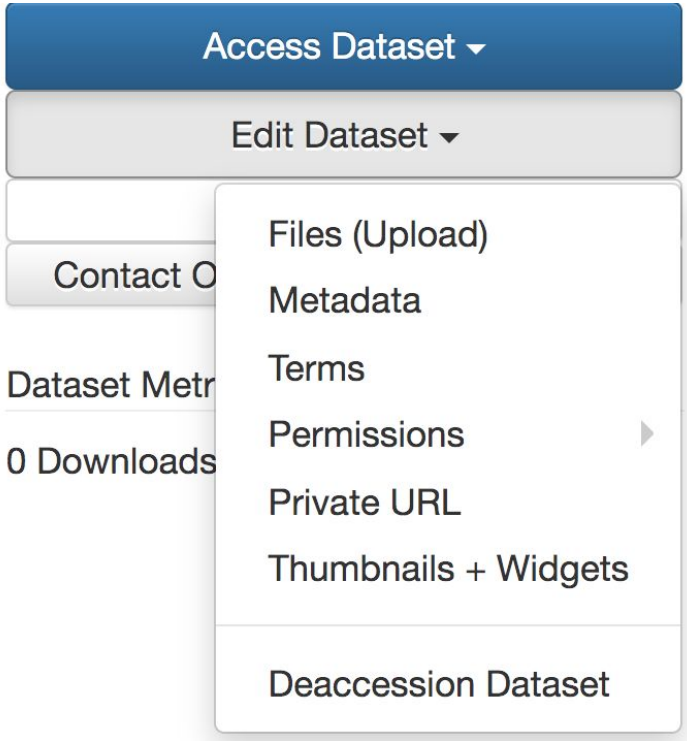
Datasets

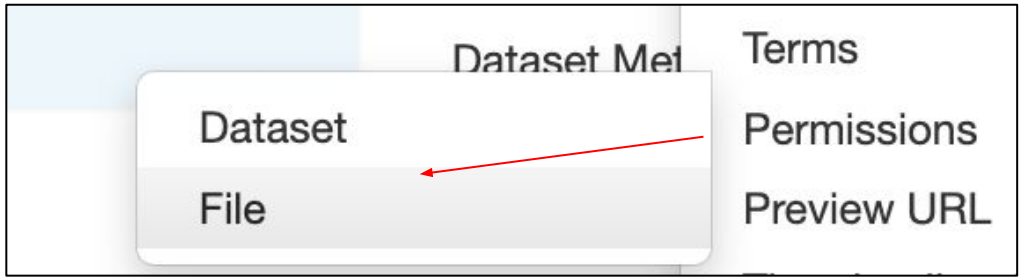
- Citation
- Metadata
- Versioning
- Private URL
- Custom Terms/Multiple License/Permissions
- Guestbooks
- Publishing Workflows

- Private URL
- Anonymous peer review
- Who can publish
- Who can access files



Datasets - Editing Options

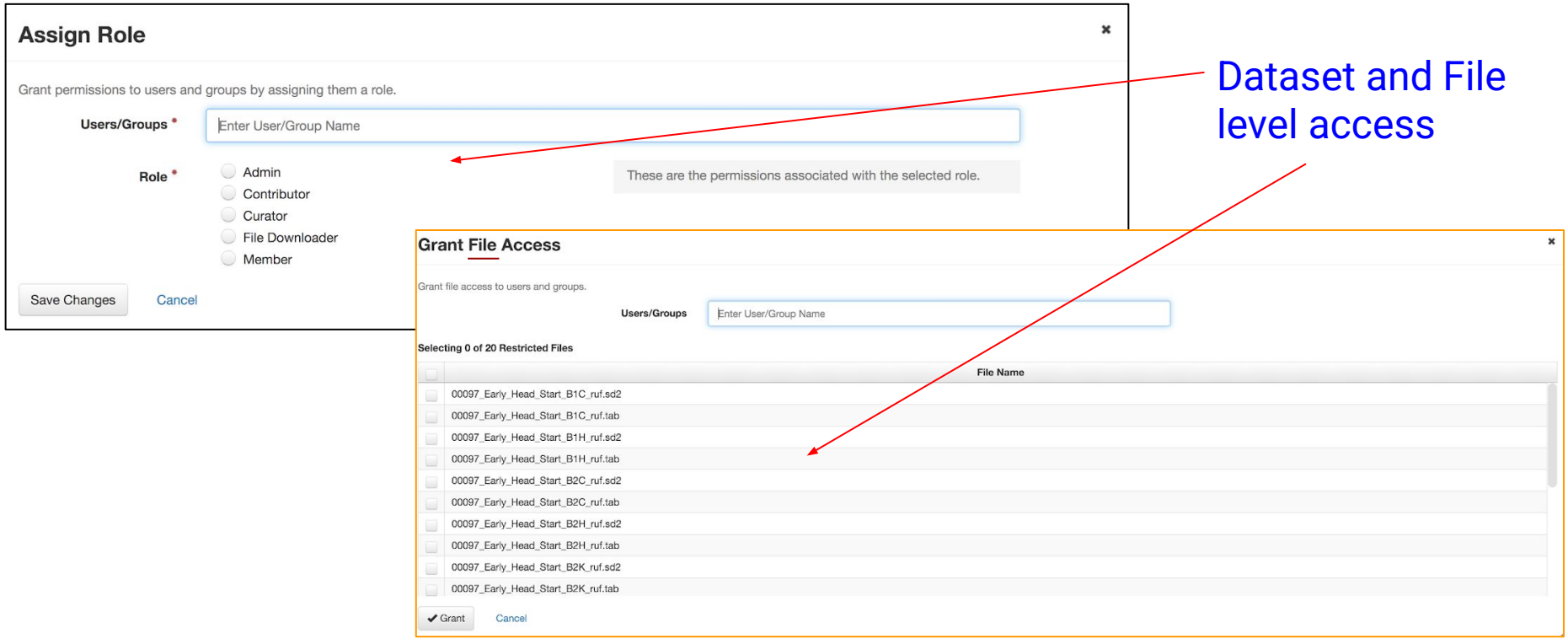




Users can receive access at two levels

- Dataset access = access to all versions
- File level access

Datasets: Permissions, Workflows and Roles

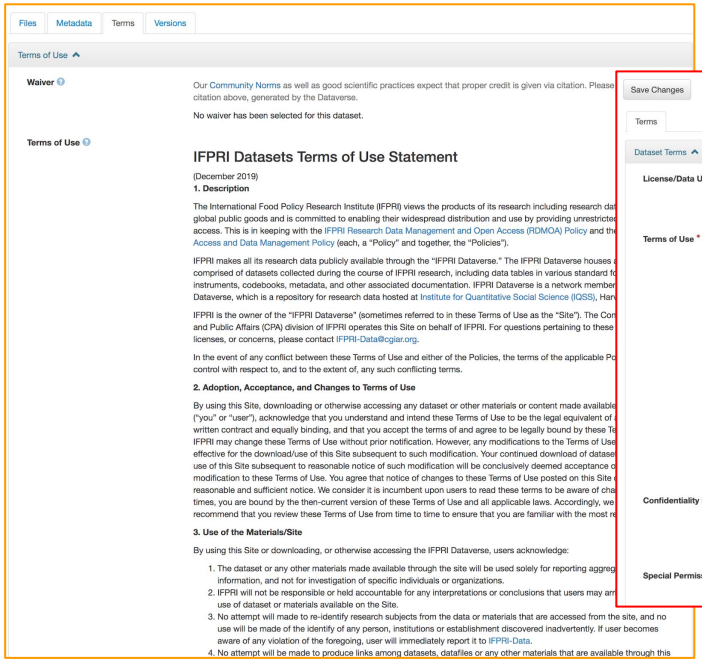


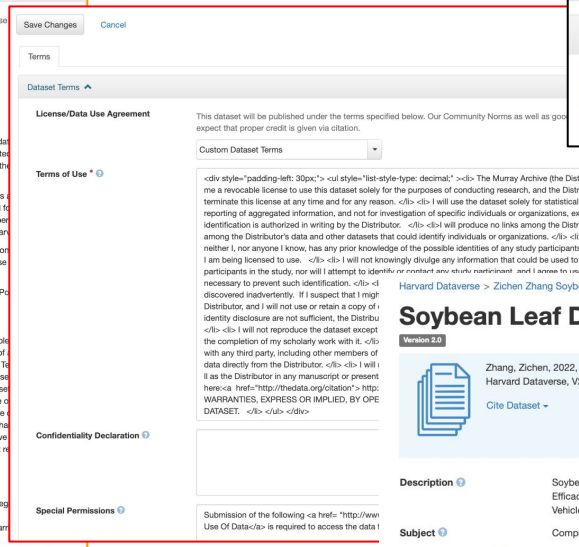
Datasets

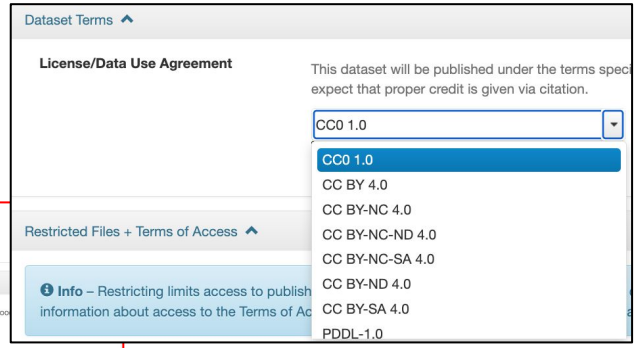


- Restricted files
- No “request access” feature enables
- See **terms of access** for instructions on data access

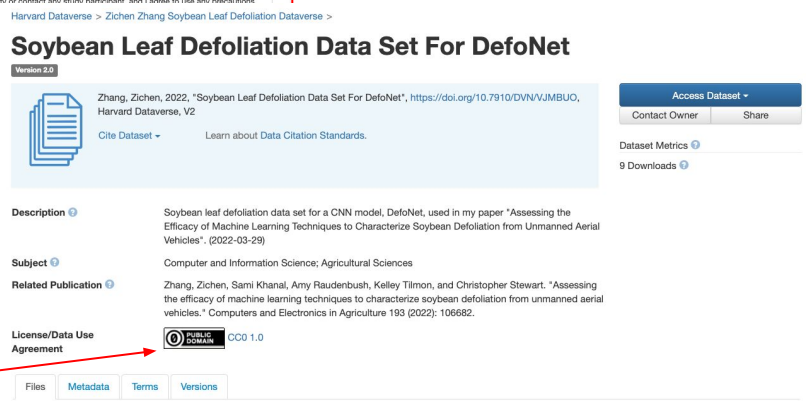
Datasets - Terms







Multiple License support



File-level

Current Features

Collections

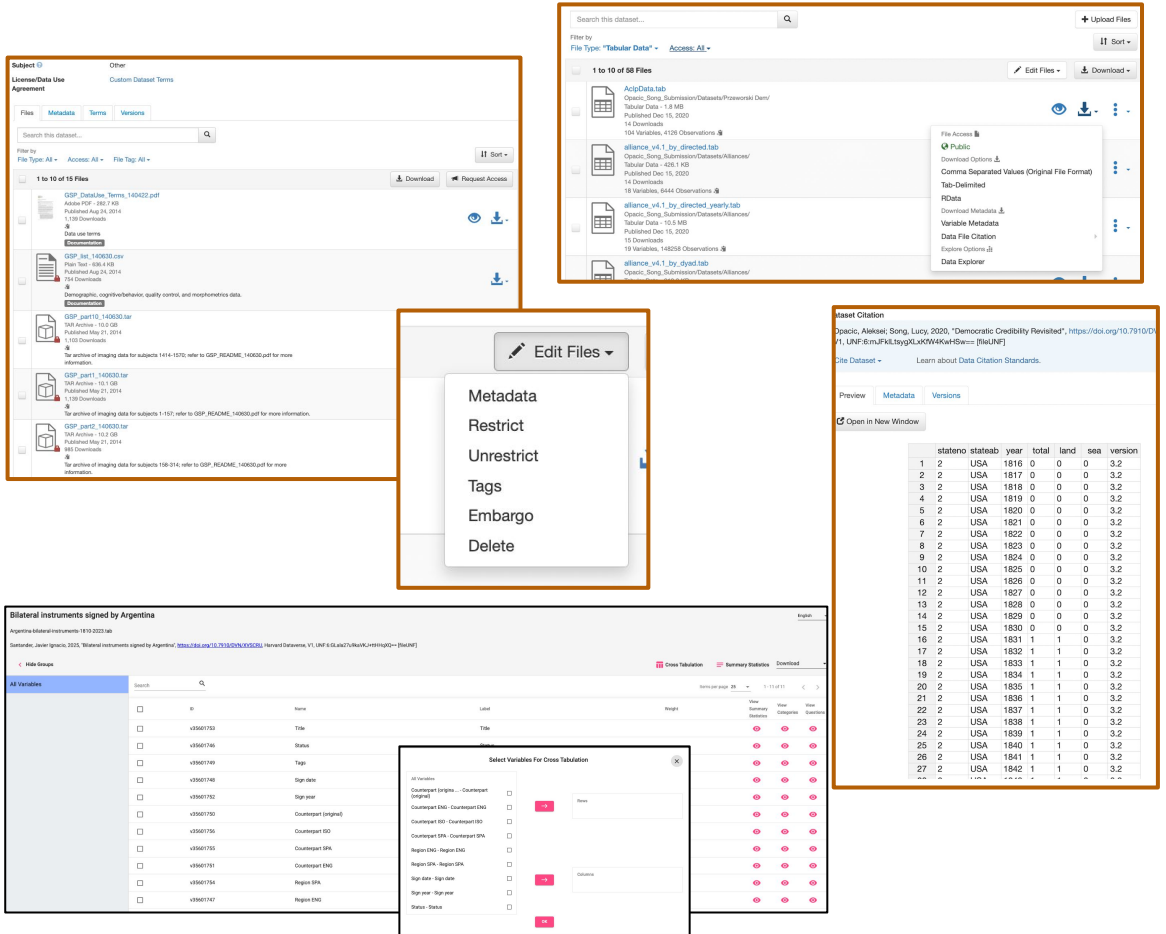
- Own administration
- Own branding (and can be embedded anywhere)

Datasets

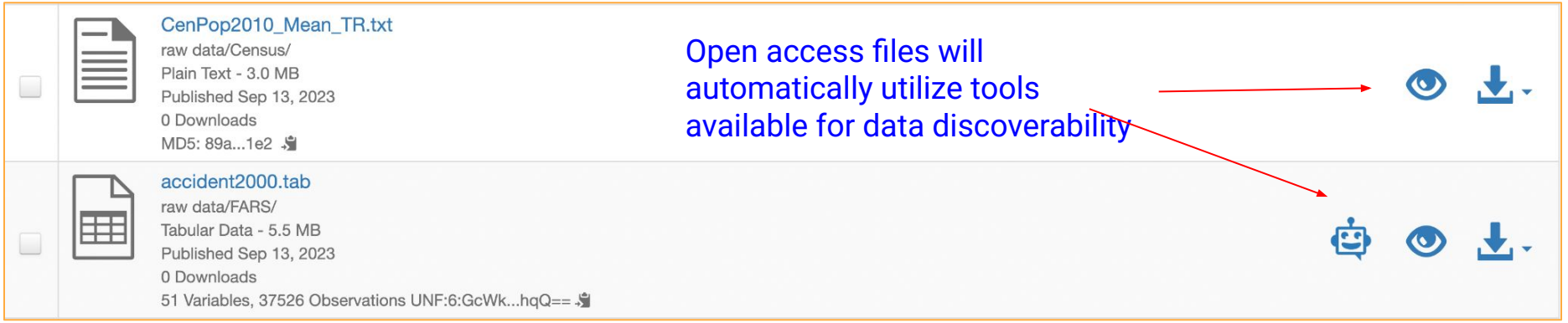
- Citation
- Metadata
- Versioning
- Private URL/*Anonymous Peer Review
- Custom Terms/*Multiple Licenses/Permissions
- Guestbooks
- Publishing Workflows

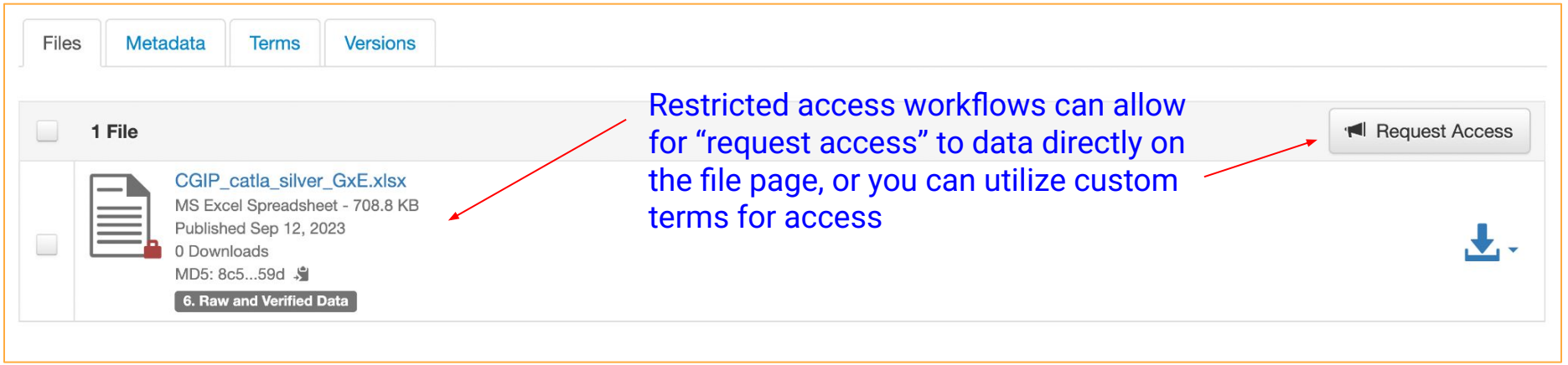
Files

- Citation
- Ingest
- Preview/Explore
- Metadata/Provenance
- Versioning
- Permissions/Embargo/Restrictions

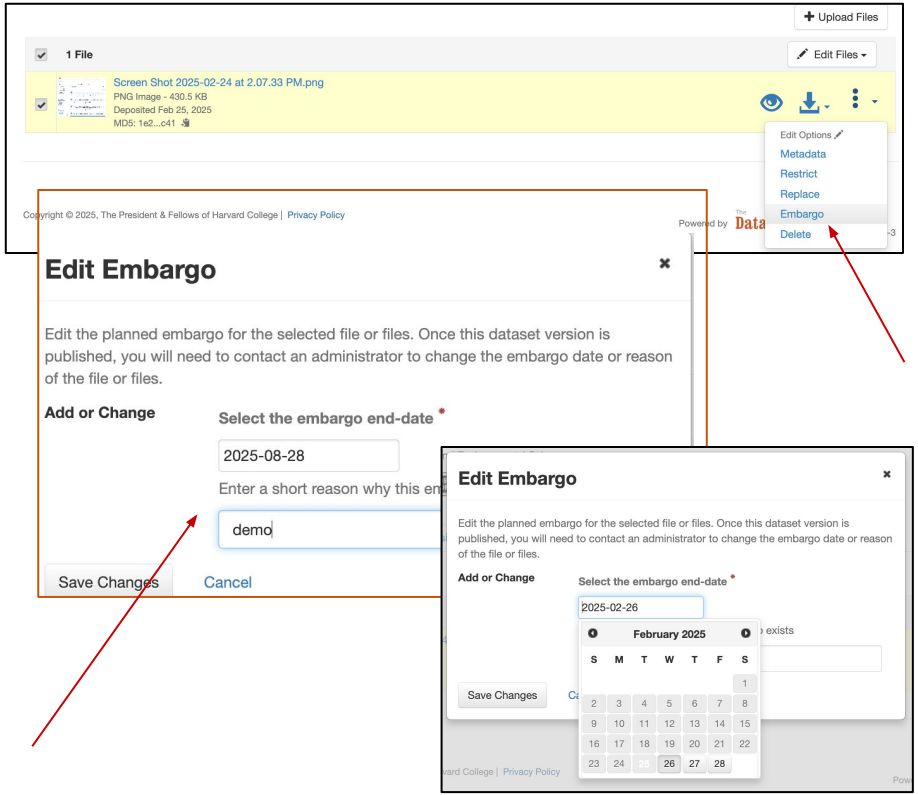


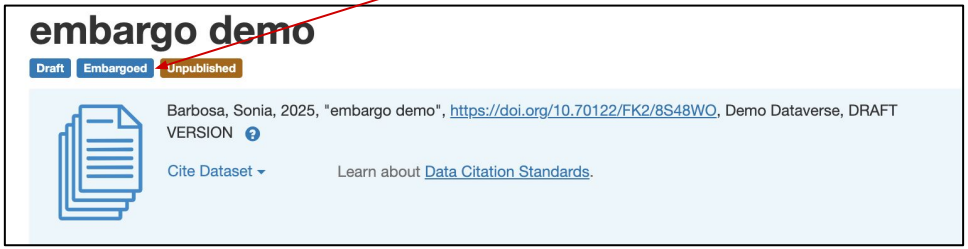
Datasets - Files

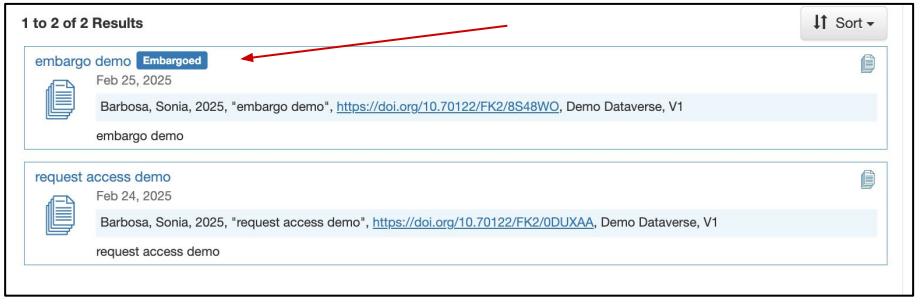


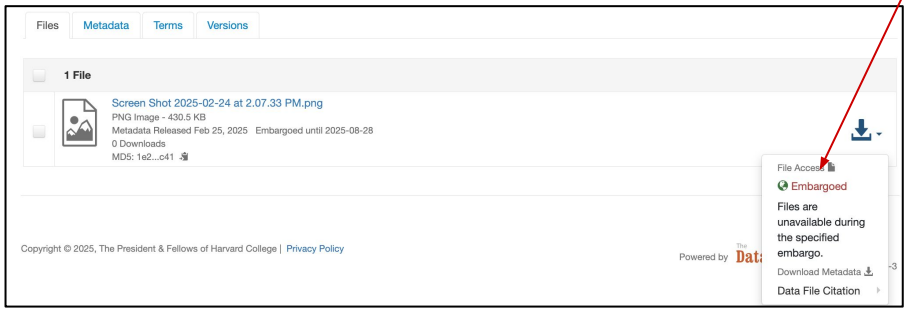


Datasets - Files







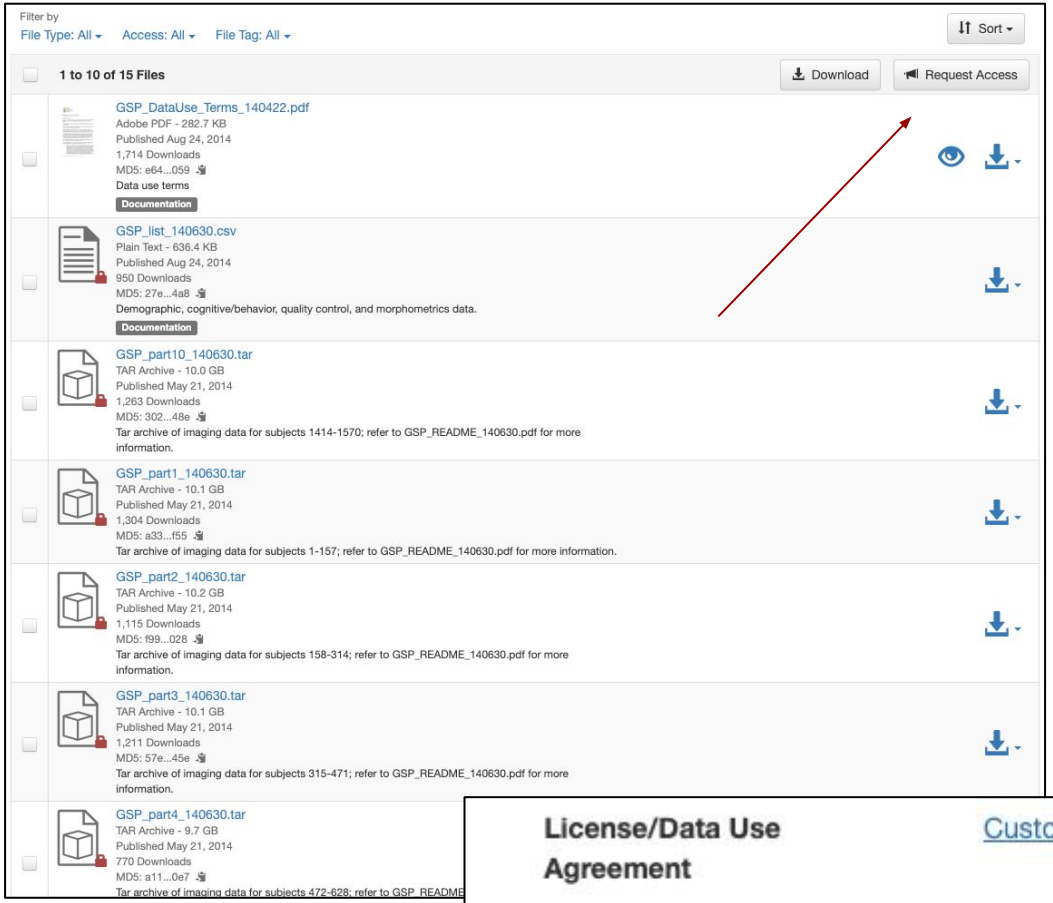


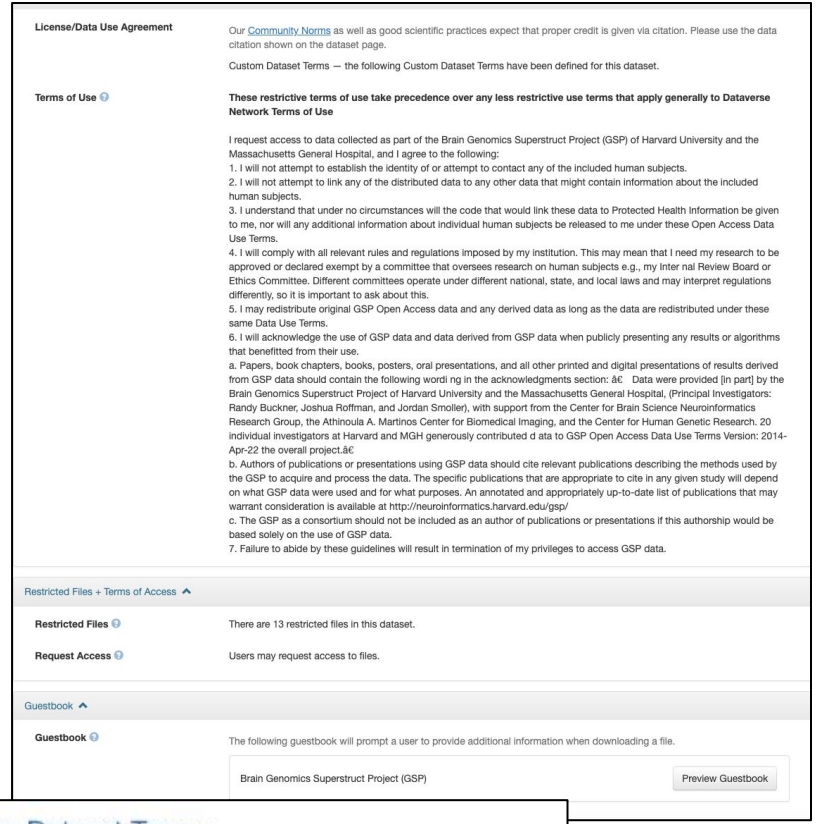
Datasets - Files



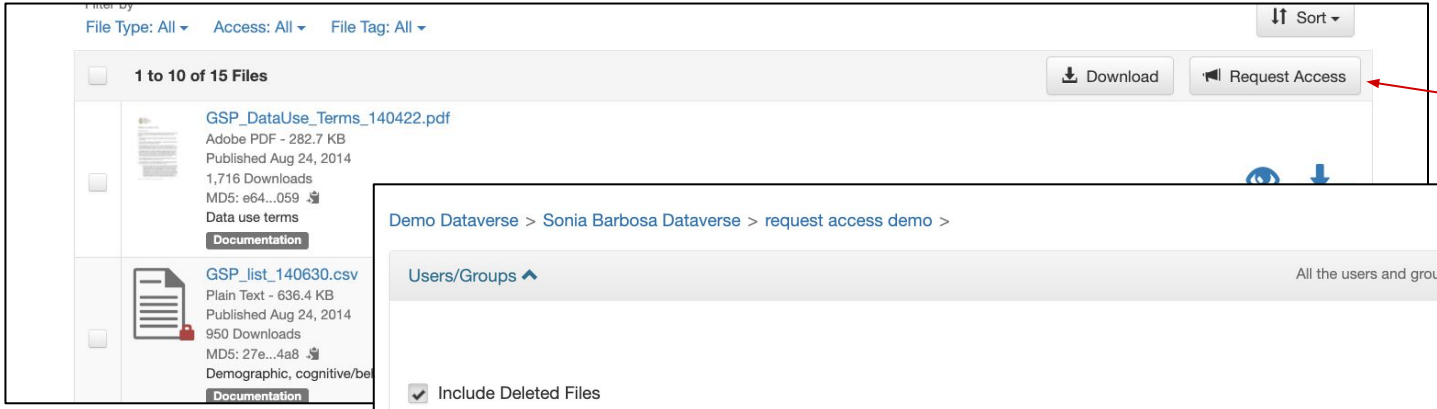
Embargo to restricted workflow

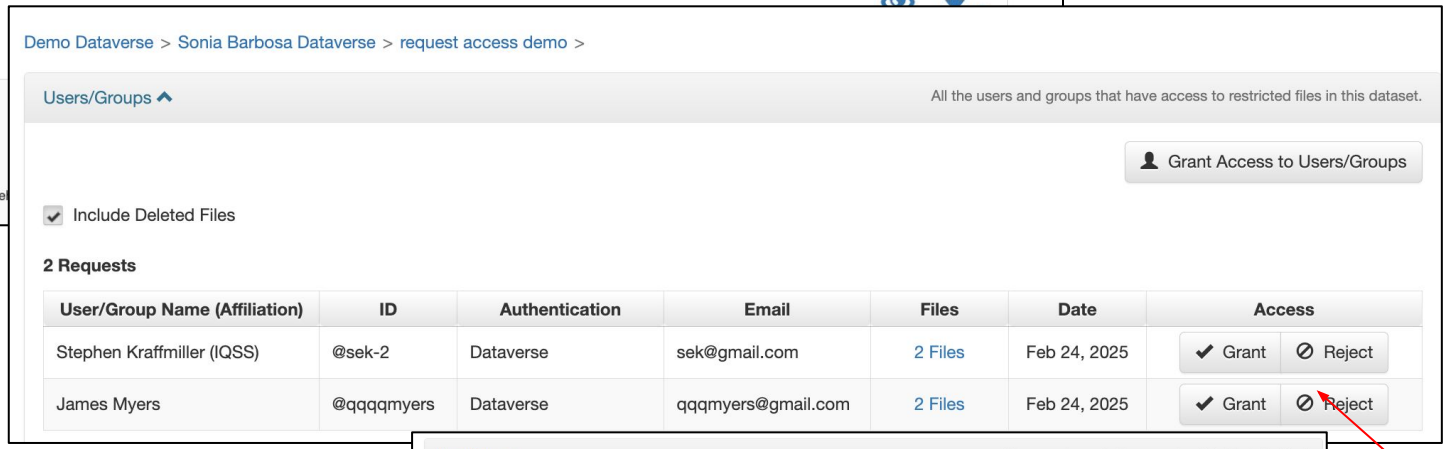
Examples of Managed Access on Harvard Repository


Filter by
File Type: All
Access: All
File Tag: All
Sort
1 to 10 of 15 Files
Download
Request Access
GSP_DataUse_Terms_140422.pdf
Adobe PDF - 282.7 KB
Published Aug 24, 2014
1,714 Downloads
MD5: e64...059
Data use terms
Documentation
GSP_list_140630.csv
Plain Text - 636.4 KB
Published Aug 24, 2014
950 Downloads
MD5: 27e...4a8
Demographic, cognitive/behavior, quality control, and morphometrics data.
Documentation
GSP_part10_140630.tar
TAR Archive - 10.0 GB
Published May 21, 2014
1,263 Downloads
MD5: 302...48e
Tar archive of imaging data for subjects 1414-1570; refer to GSP_README_140630.pdf for more information.
GSP_part1_140630.tar
TAR Archive - 10.1 GB
Published May 21, 2014
1,304 Downloads
MD5: a33...f55
Tar archive of imaging data for subjects 1-157; refer to GSP_README_140630.pdf for more information.
GSP_part2_140630.tar
TAR Archive - 10.2 GB
Published May 21, 2014
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These restrictive terms of use take precedence over any less restrictive use terms that apply generally to Dataverse Network Terms of Use
I request access to data collected as part of the Brain Genomics Superstruct Project (GSP) of Harvard University and the Massachusetts General Hospital, and I agree to the following:
1. I will not attempt to establish the identity of or attempt to contact any of the included human subjects.
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a. Papers, book chapters, books, posters, oral presentations, and all other printed and digital presentations of results derived from GSP data should contain the following wordi ng in the acknowledgments section: â€ Data were provided [in part] by the Brain Genomics Superstruct Project of Harvard University and the Massachusetts General Hospital, (Principal Investigators: Randy Buckner, Joshua Roffman, and Jordan Smoller), with support from the Center for Brain Science Neuroinformatics Research Group, the Athinoula A. Martinos Center for Biomedical Imaging, and the Center for Human Genetic Research. 20 individual investigators at Harvard and MGH generously contributed d ata to GSP Open Access Data Use Terms Version: 2014-Apr-22 the overall project.â€
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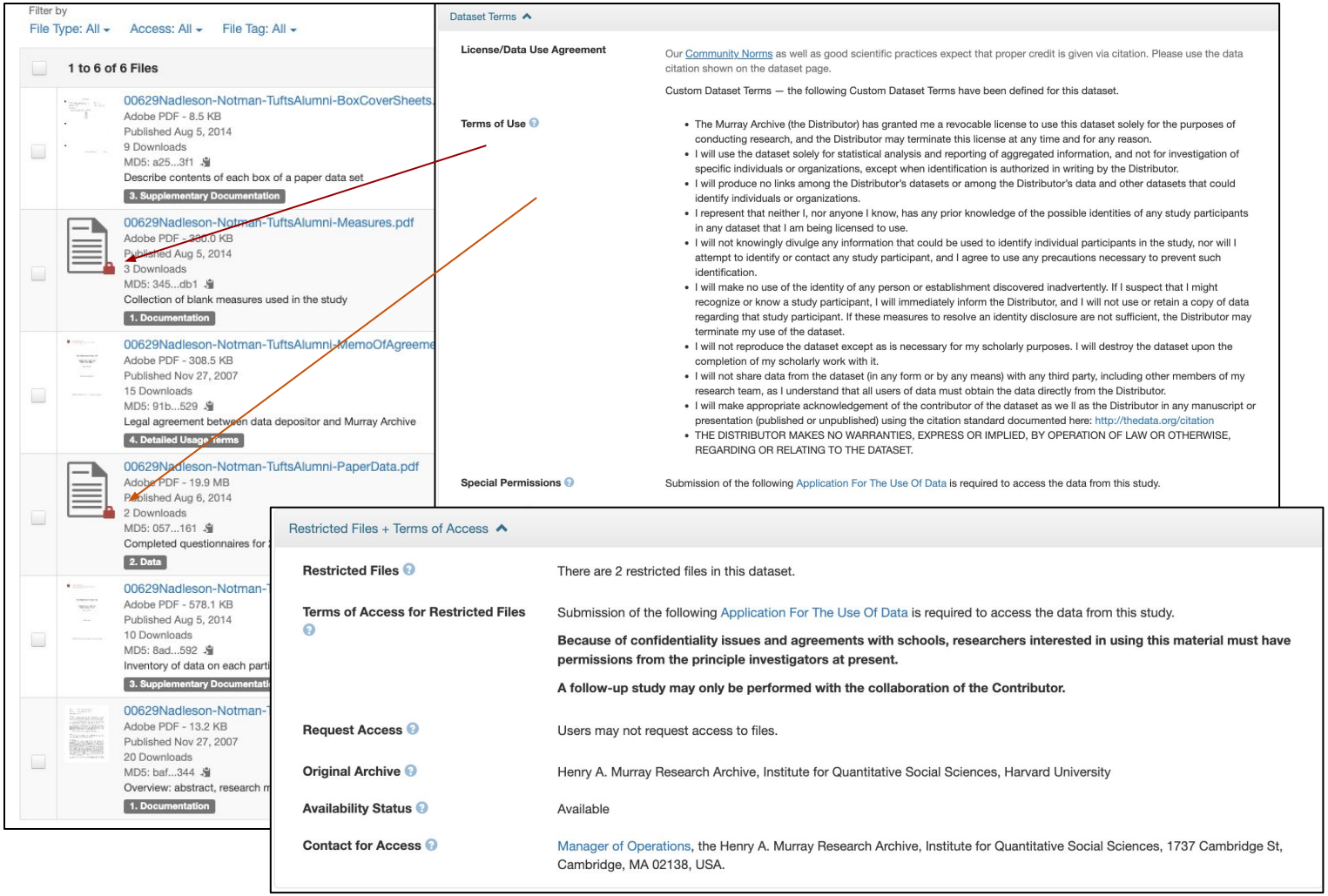

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- Anonymized human subjects collection on Harvard Repository
- Restricted data files, custom license and application form
- Requires that collection admins assign the file level or dataset access level once the requester has created their account
- Involved more administrative oversight



- Embargoes to open access
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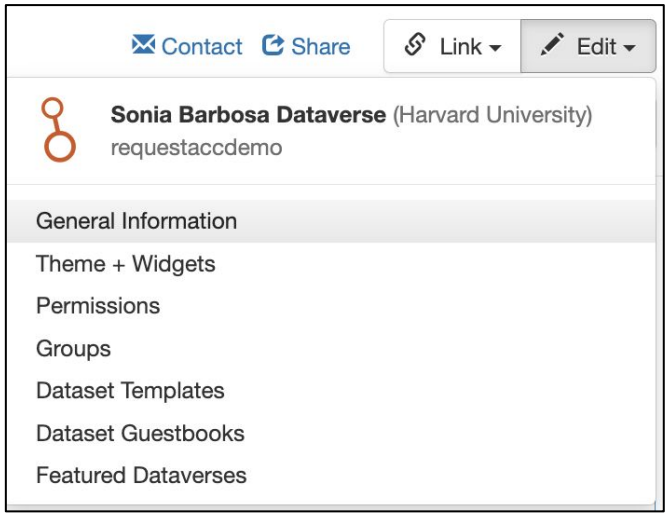
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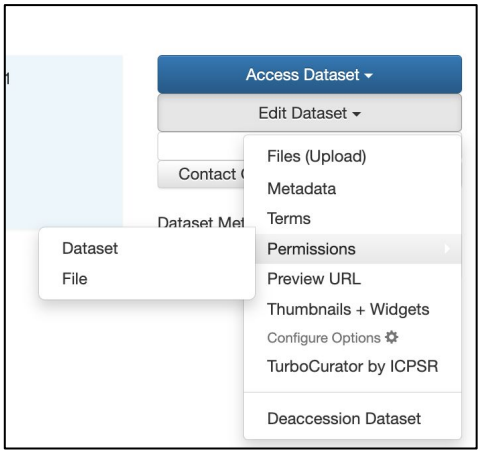
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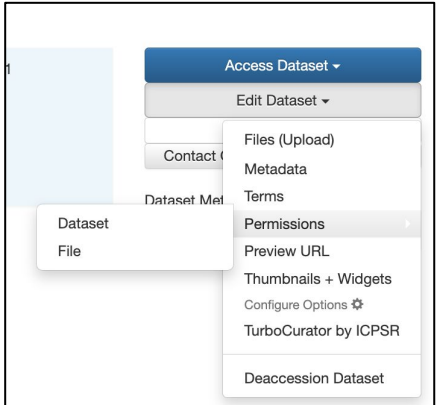
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References

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